

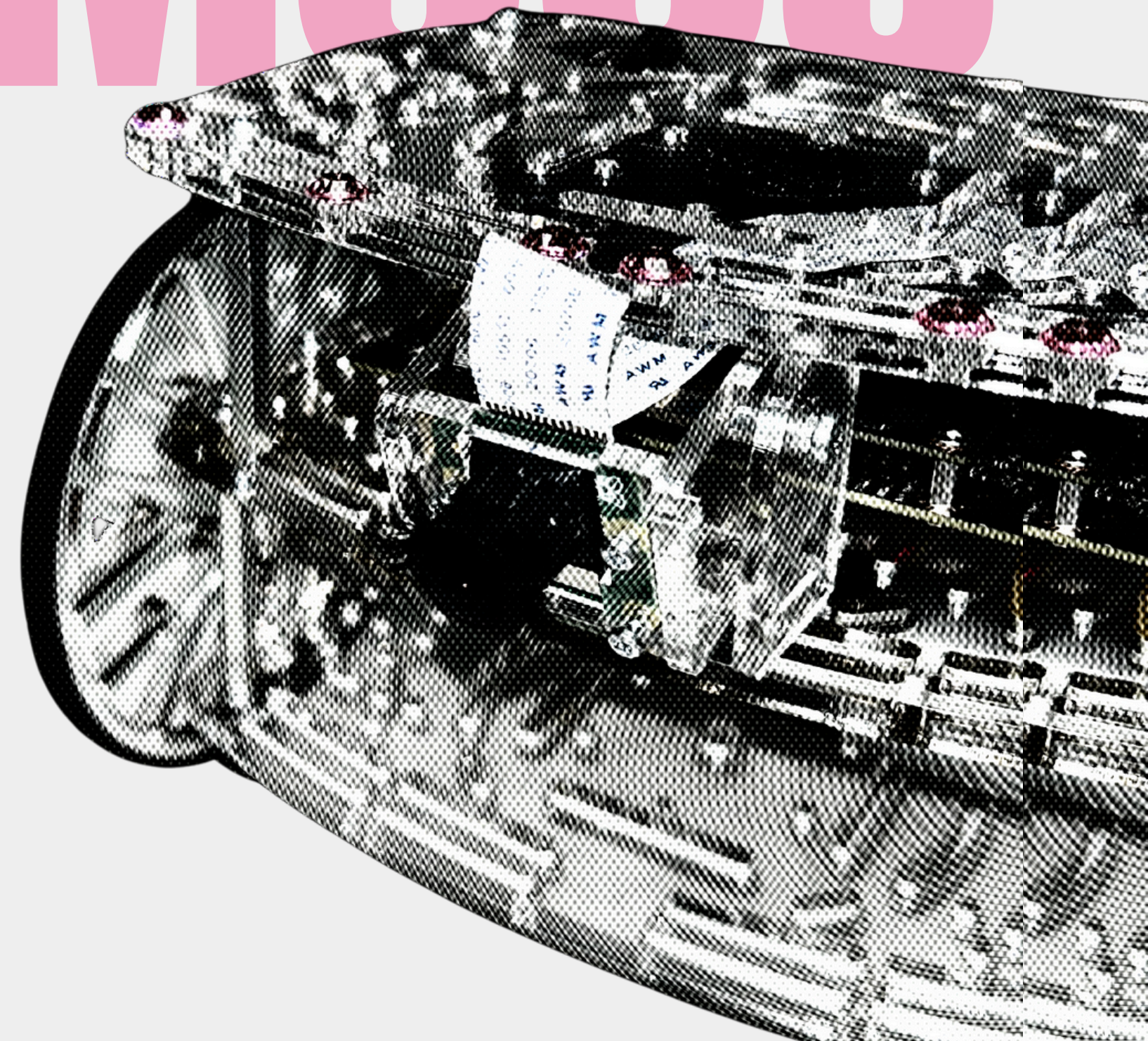
FINAL CHALLENGE

JUNE/2025

AUTONOMOUS DRIVING

Team 2

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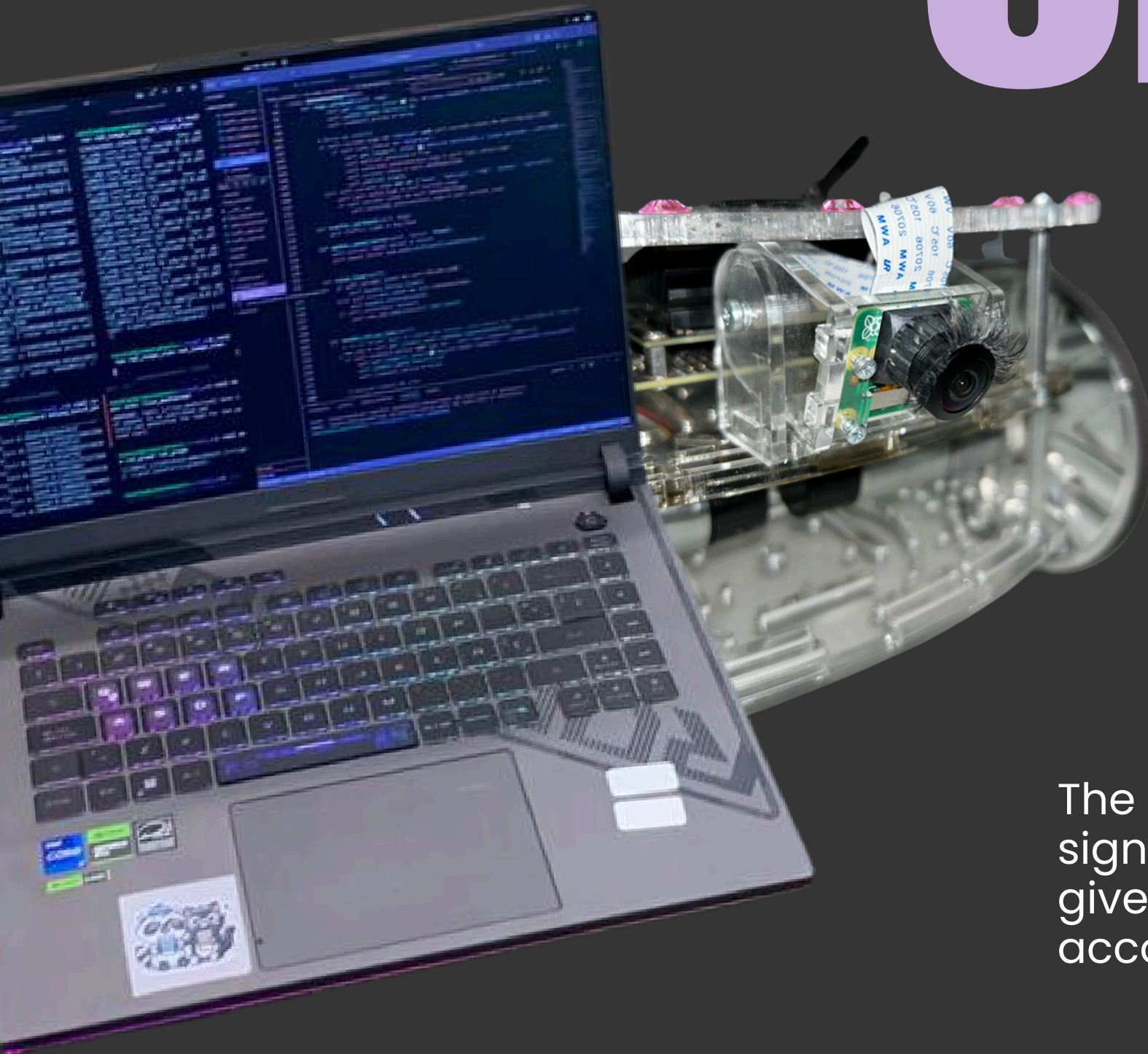


INTRODUCTION

Autonomous navigation in robotics allows the robot to move without human intervention, by processing information from the environment.

Detection of traffic signals and traffic lights is essential to ensure safe behavior in accordance with road rules, especially in urban scenarios.

OBJECTIVES



Trajectory

The robot completes the entire trajectory following the black line, respecting traffic signs and traffic lights.

Traffic signals

The robot correctly detects each signal (forward, turn, road work, give way) and responds accordingly.

Traffic light

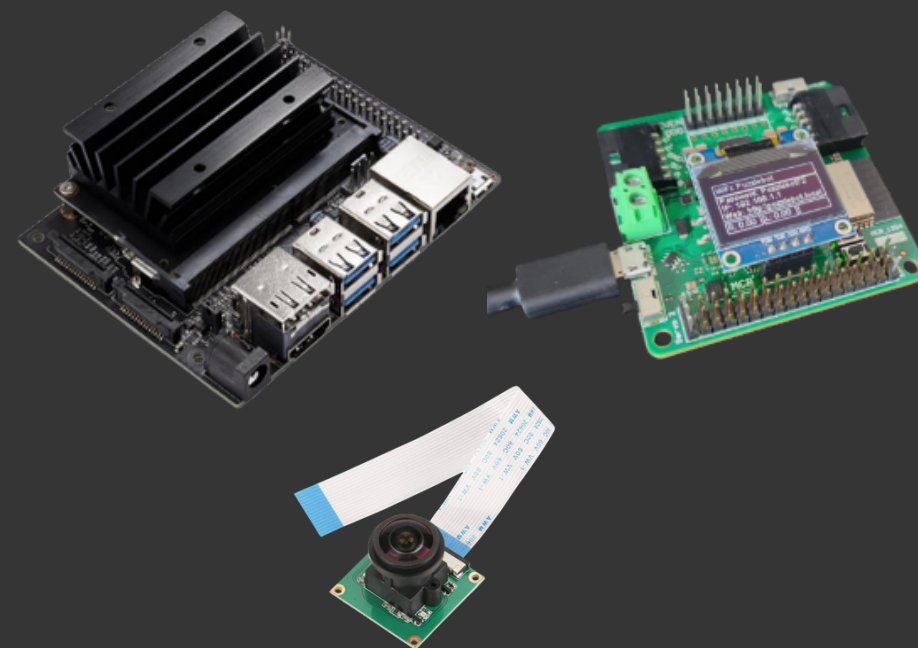
The robot reacts autonomously and correctly to the traffic light colors (red, yellow, green).

PHYSICAL AND ELECTRONIC DESIGN



MCR Kit

- NVIDIA Jetson Nano
- Hackerboard
- Raspberry Pi Camera



Additional

- Batteries
- Passive Buzzer
- DC Brushless Fan

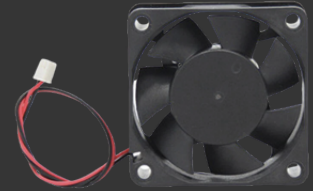
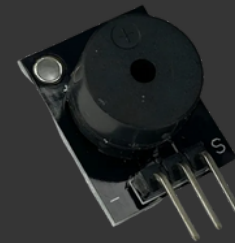


Additional



- Batteries → to correctly power the hackerboard and the jetson at the same time.

- Passive Buzzer → to activate when passing by specific signals.

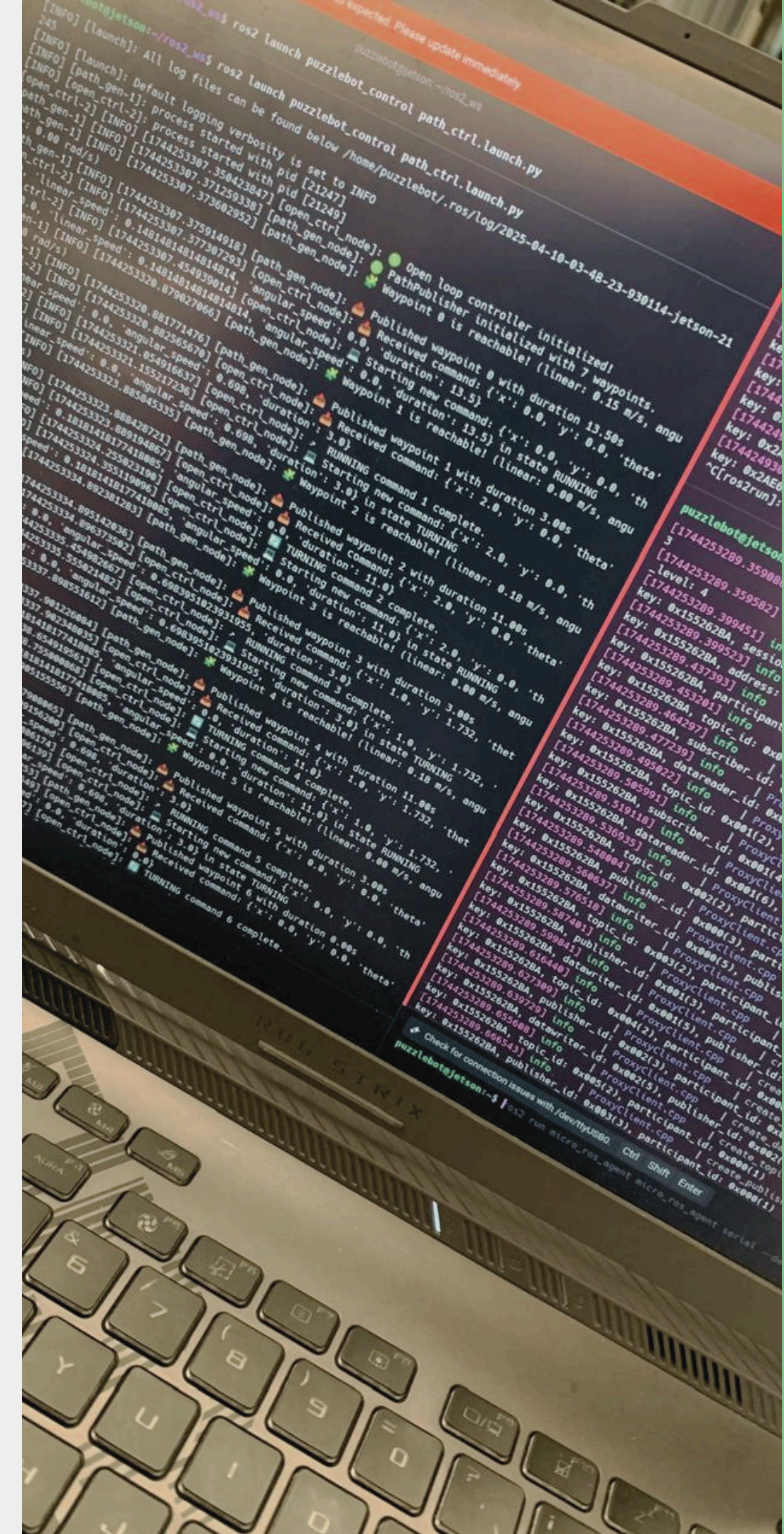


- DC Brushless Fan → to prevent Jetson from overheating.



SOFTWARE ENGINEERING STRATEGY

- 5 ROS Nodes
- 2 Trained YOLO MODELS



4 ROS2 Nodes

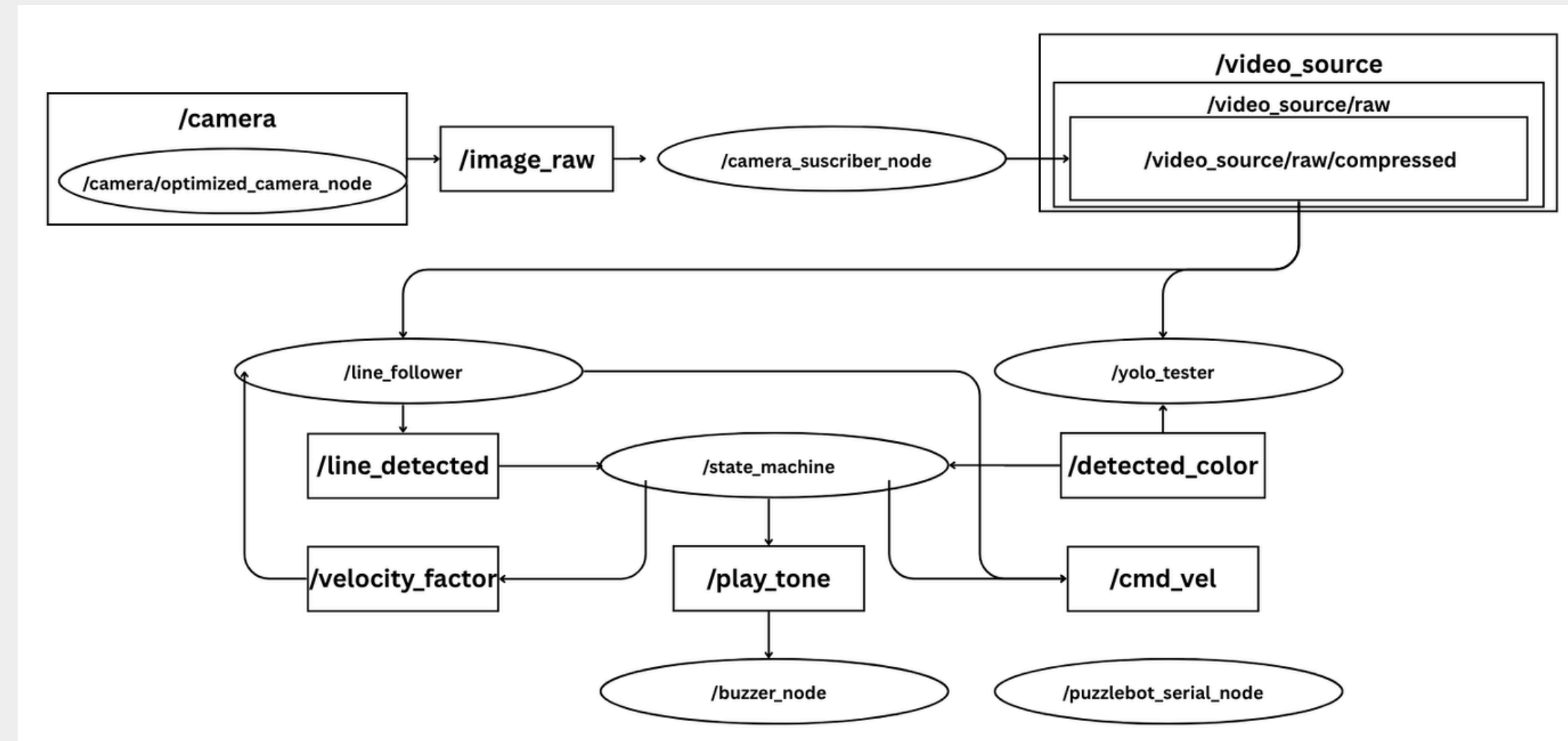
1. State Machine

2. Line Follower

3. Buzzer

4. Traffic Signs

5. Traffic Light



State Machine

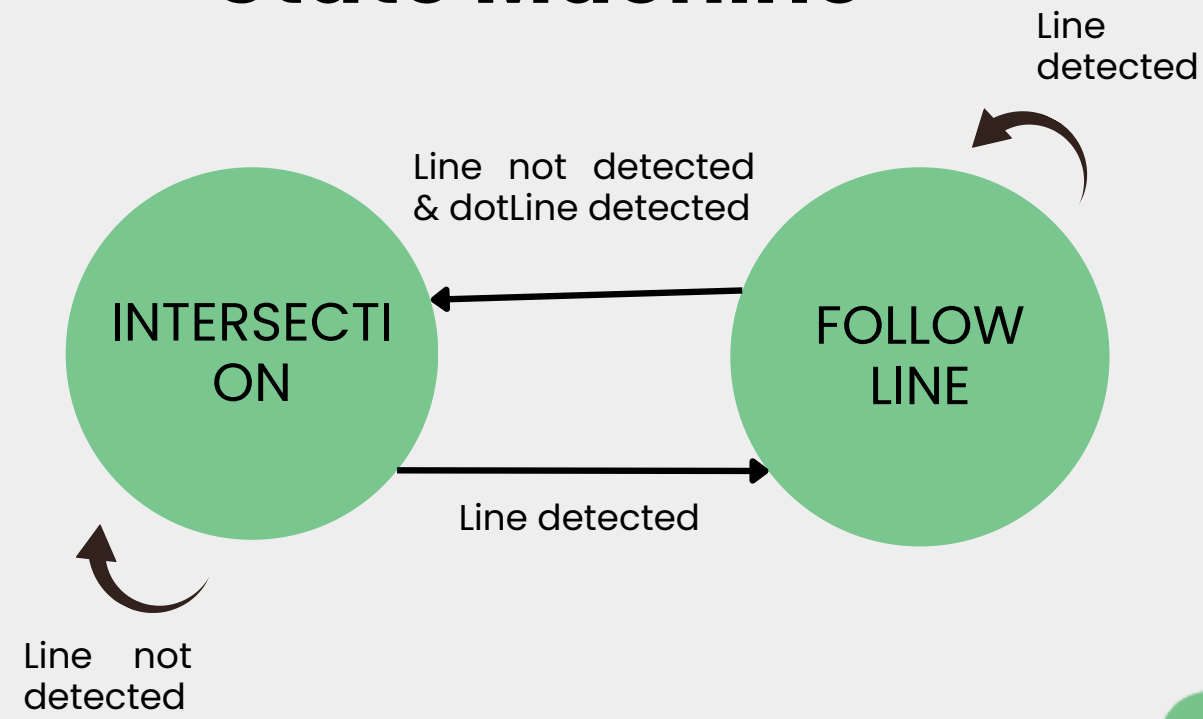
Controls autonomous navigation with two states:

- FOLLOW_LINE: follows the line (/line_detected) and reacts to traffic signs
- INTERSECTION: reacts to traffic signs at intersection (/detected_sign) and lights (/detected_color)

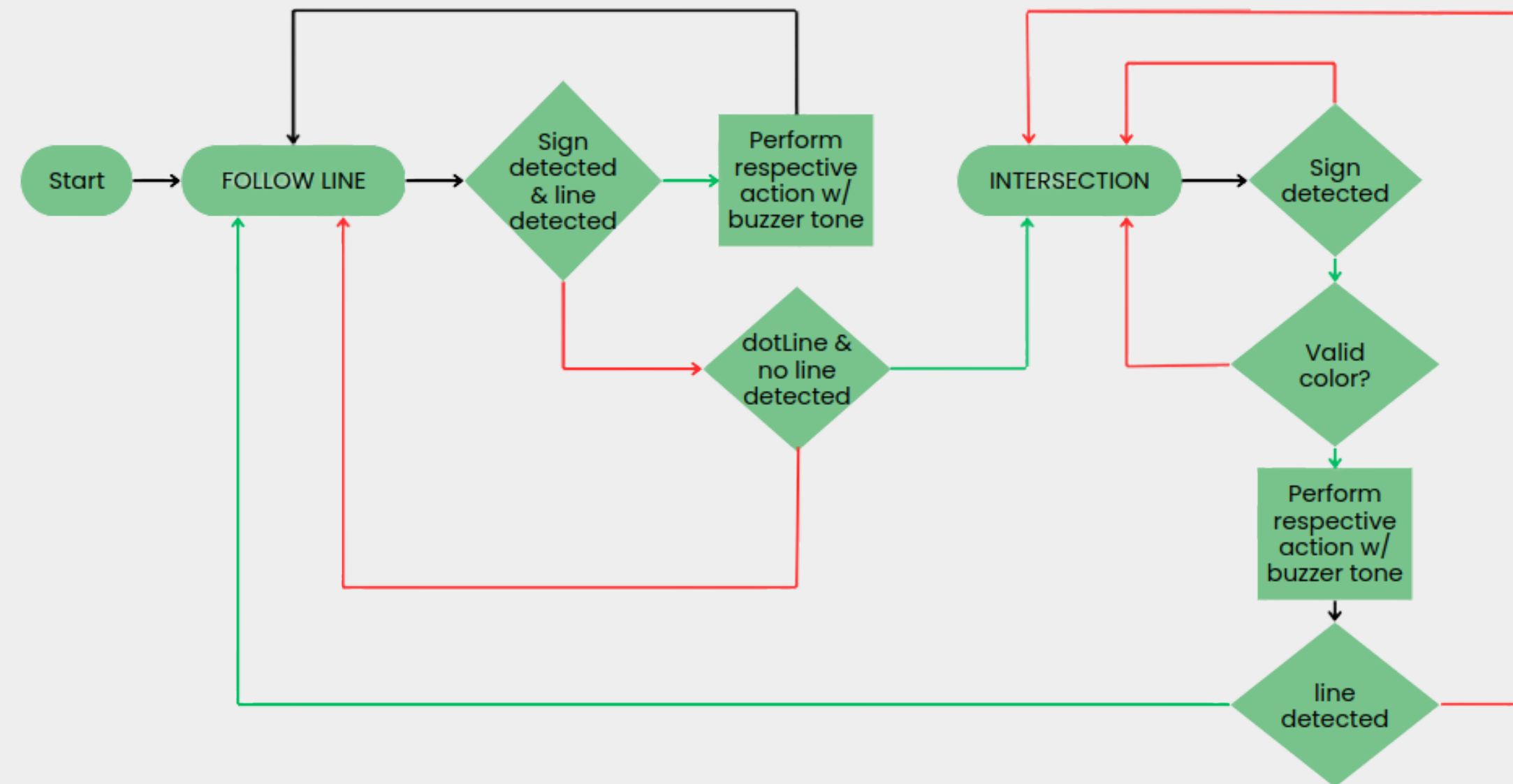
Handles signs like stop, giveWay, roadwork, aheadOnly, turnLeft, and turnRight.

Publishes motion (/cmd_vel), buzzer sound (/play_tone), and speed factor (/velocity_factor) for the line_follower node.

State Machine

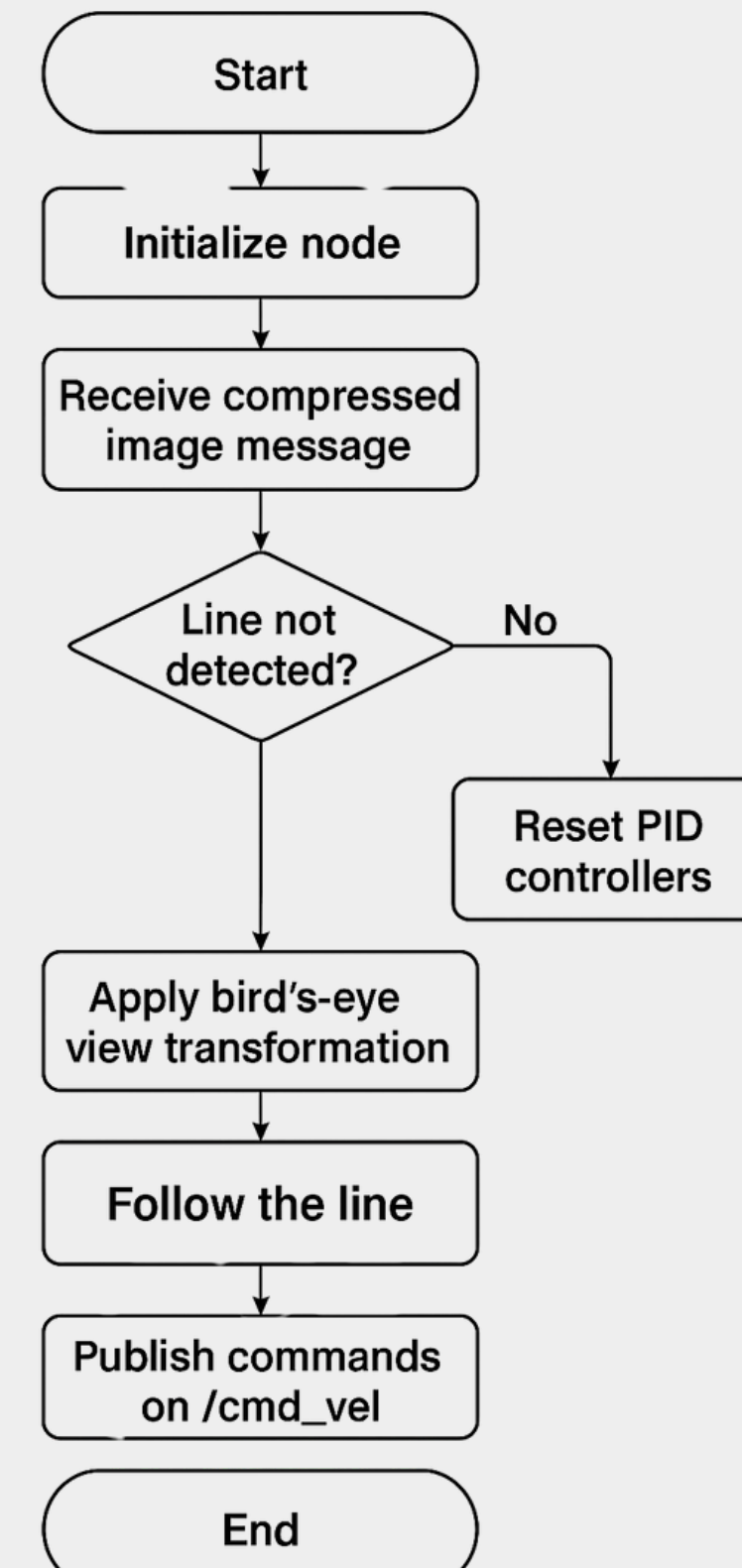


Flowchart



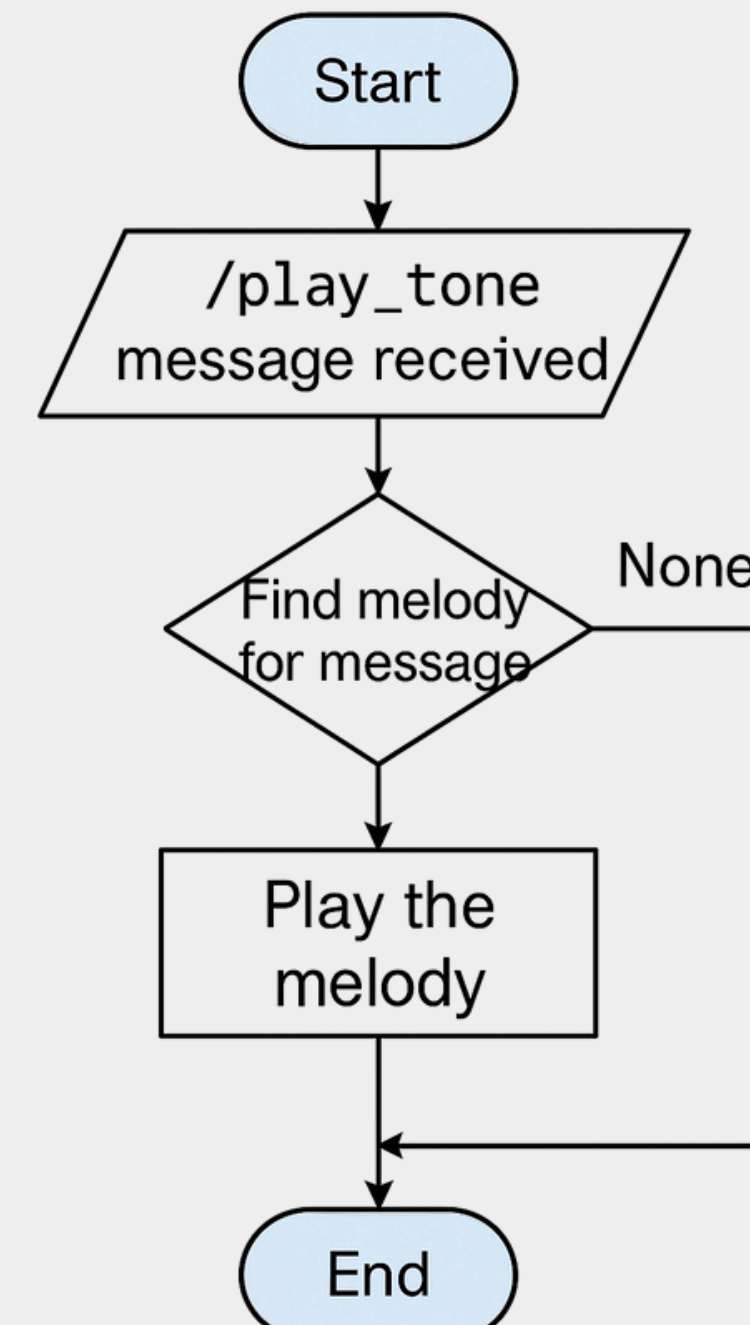
Line Follower

Detects and follows a line using compressed images and PID control. It adjusts speed and turns according to straights or curves, publishes to `/cmd_vel` and can be activated with a service.



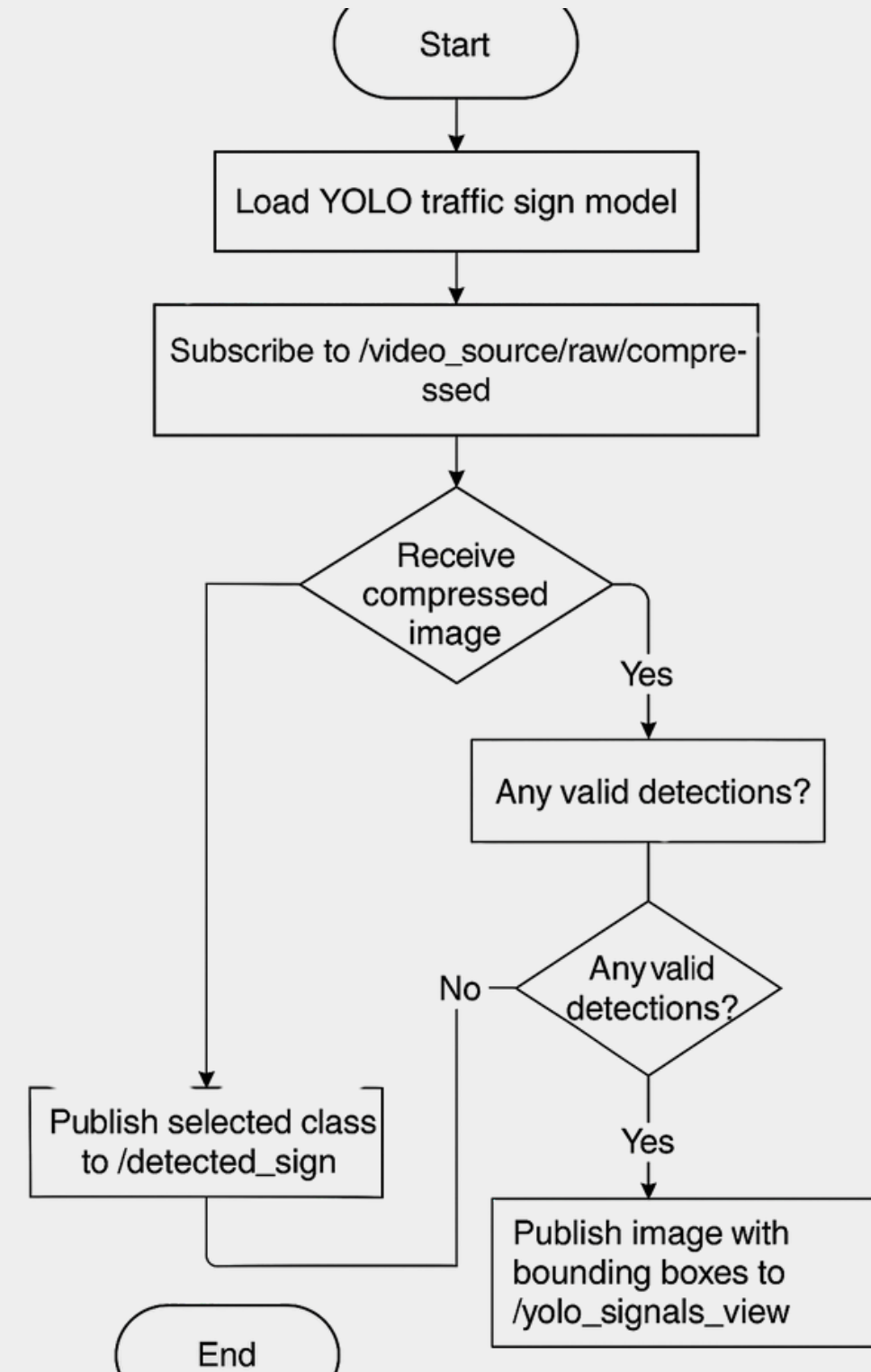
Buzzer

Receives messages in the `/play_tone` topic and runs specific tunes on a buzzer connected to the Jetson Nano via PWM. Each detected signal has its own frequency pattern and duration. The node uses the Jetson.GPIO library to control the buzzer.



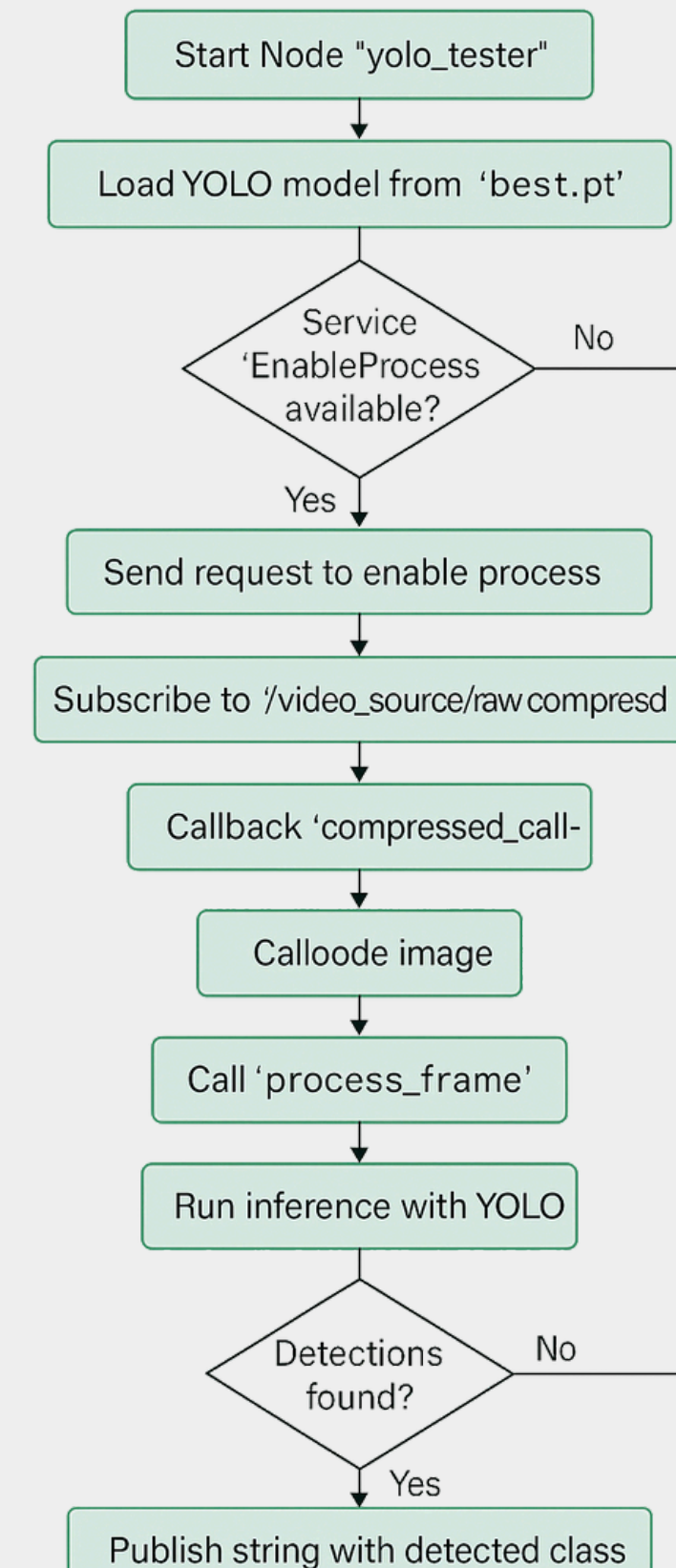
Traffic Signs

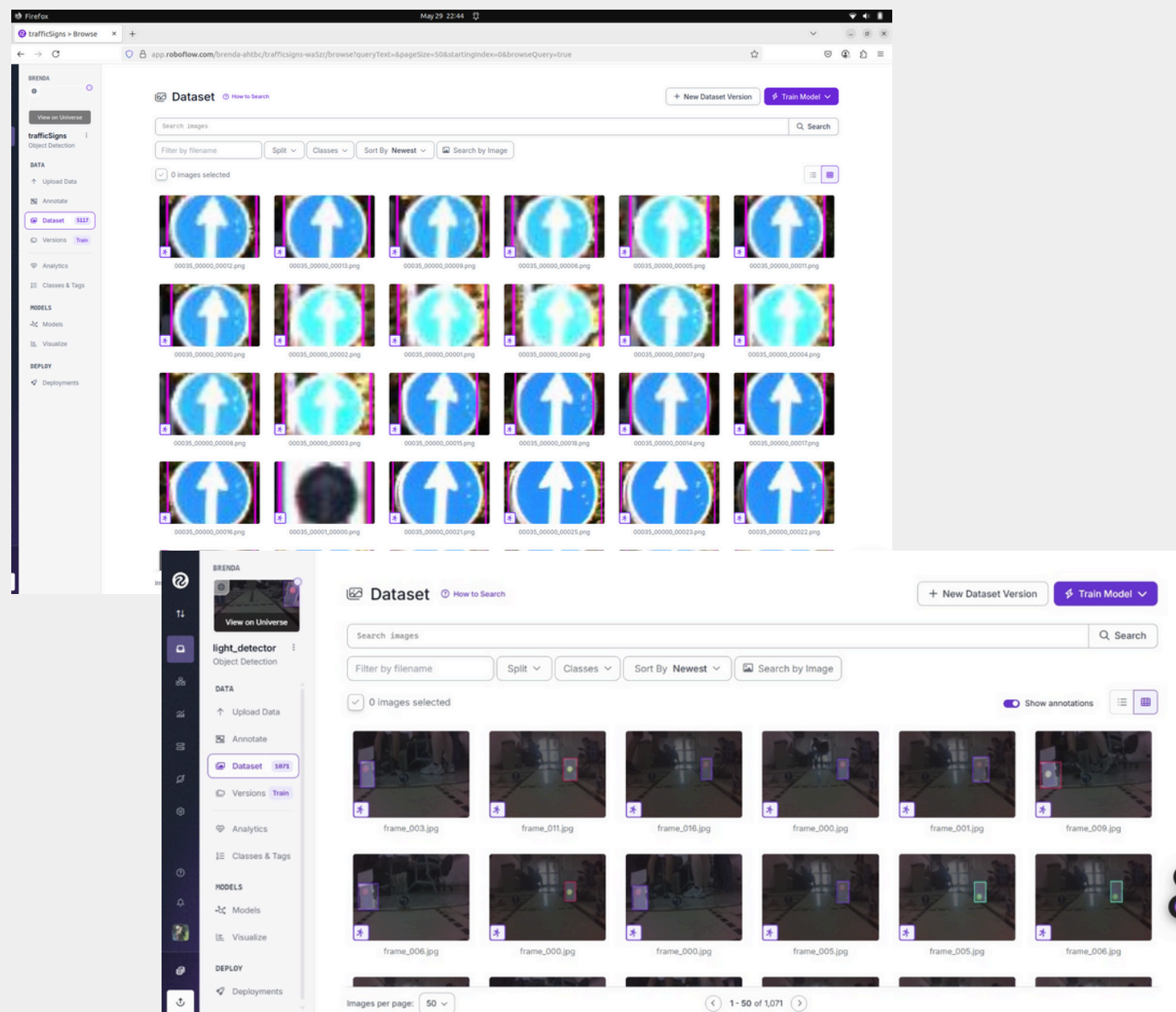
Detects traffic signs in compressed images using a custom YOLO model. Publishes the detected class in `/detected_sign` only with high confidence signals.



Traffic Light

Detects traffic light colors in compressed images using a custom YOLO model. On startup, it activates the system using an EnableProcess service. If active, it publishes the detected class (red, yellow, green) in /detected_color.



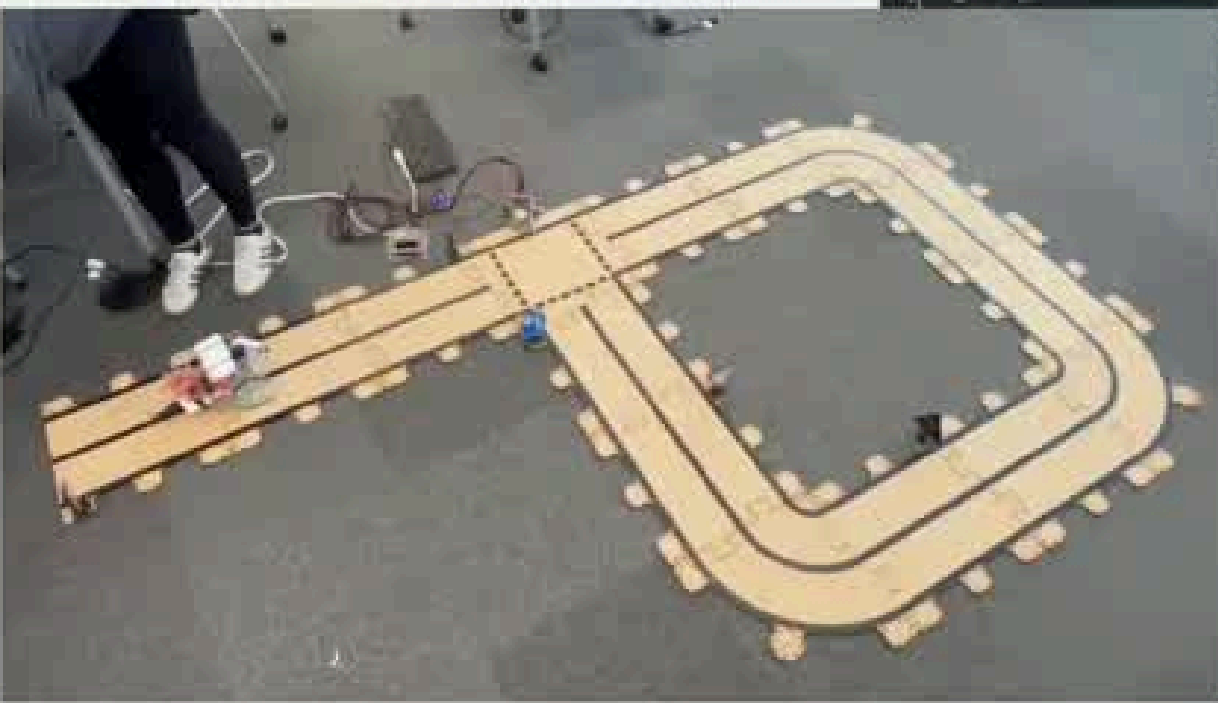
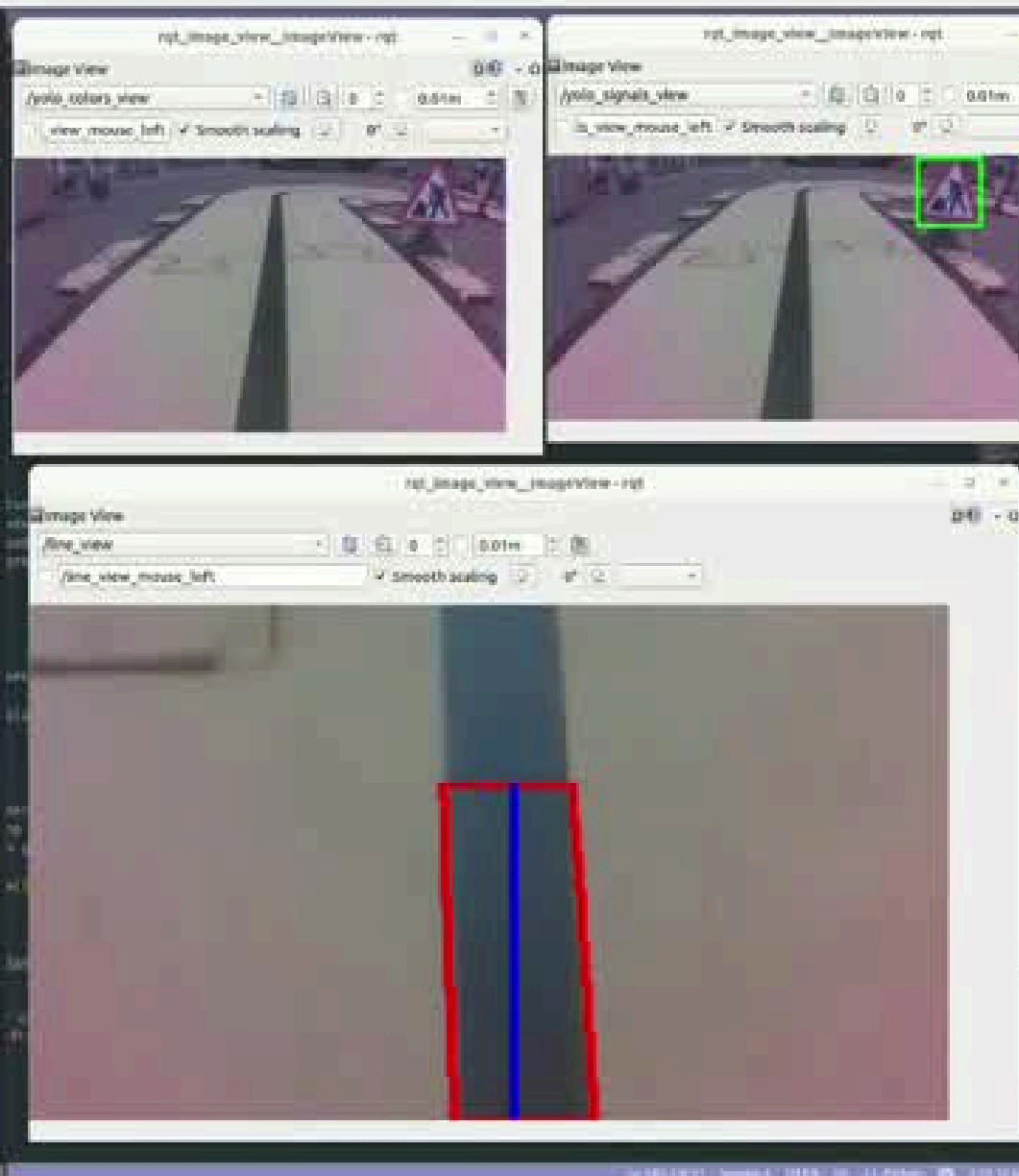
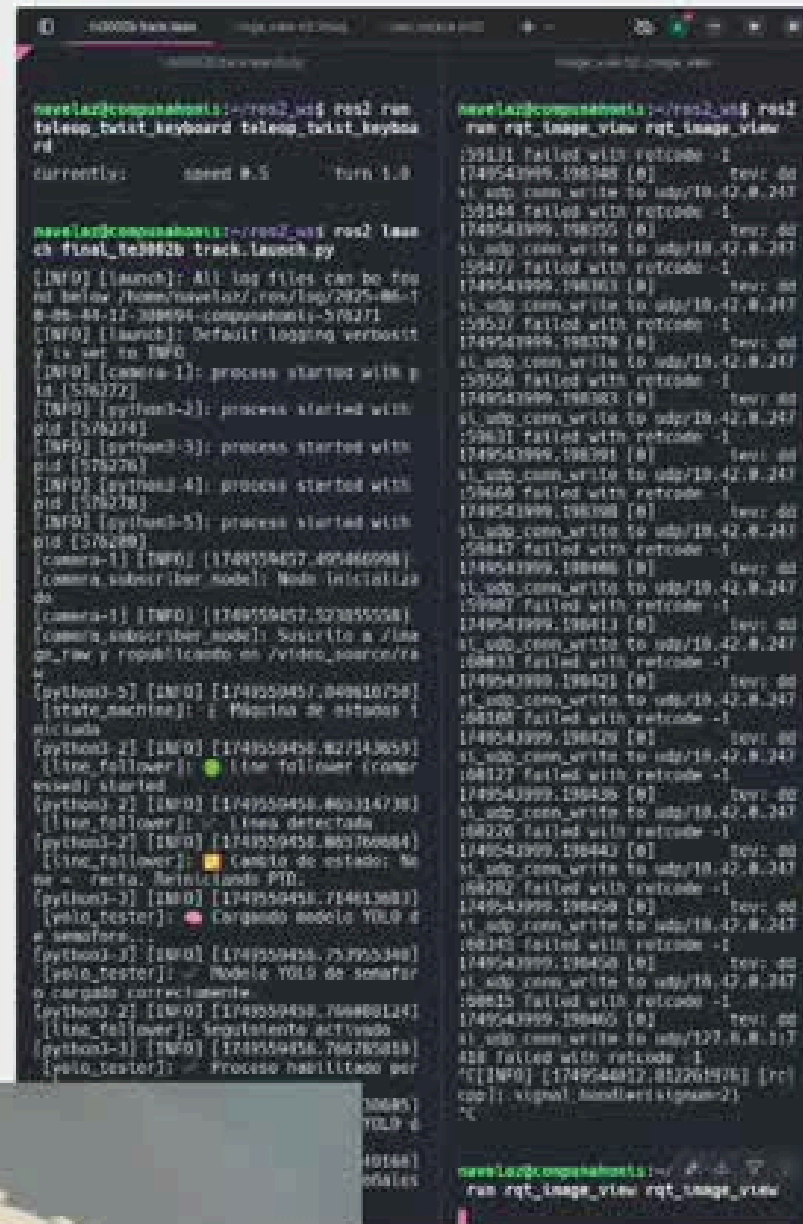
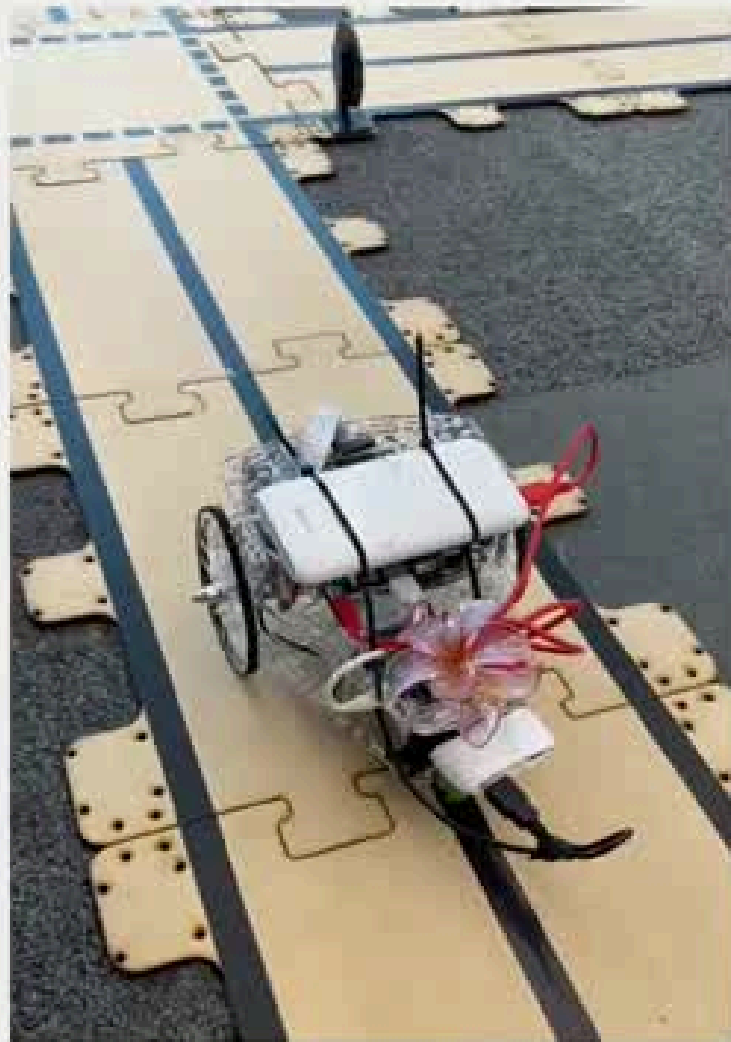


YOLO Models

Trained on Roboflow repository data and frames taken from videos recorded at the track.

Traffic Signs

Traffic Light

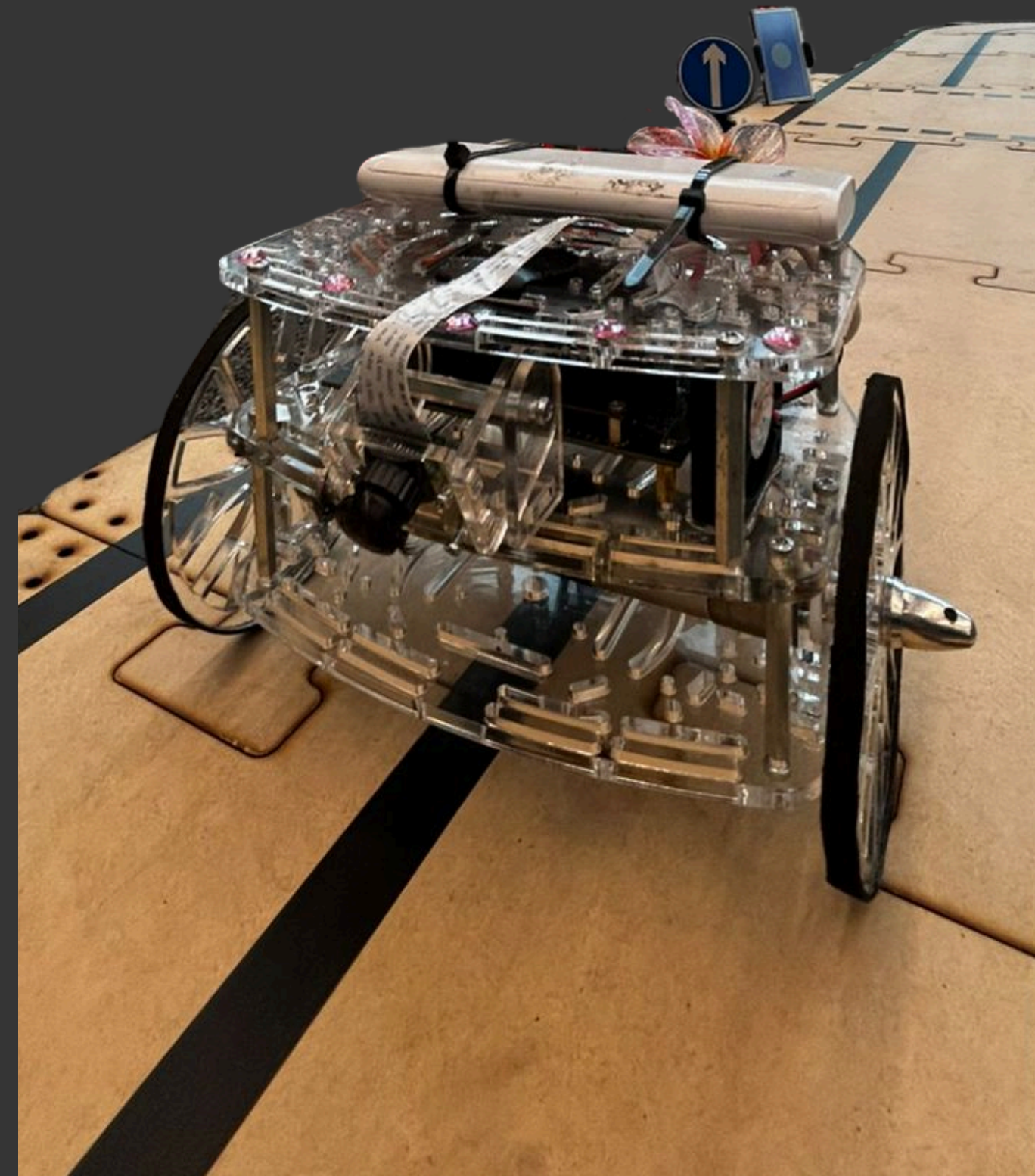


DEMO

-  Ahead only:
Continue straight ahead only.
- / Turn left/right :
Turn left or right further ahead.
-  Road work: Reduce the speed of the robot.
-  Stop: Stop the robot.
-  Give way: Give way to traffic on main roads. Slow down until you reach the intersection and yield.
- **Red:** Stop until you see the green light.
- **Yellow:** Drive slowly.
- **Green:** Continue on your route.

CONCLUSION & FUTURE WORK

The system enabled autonomous navigation of a differential robot using both line following and vision-based detection of traffic signs and lights. Mechanical issues with motor alignment and mounting were resolved through iterative adjustments, highlighting the importance of robust hardware-software integration in autonomous robotics.





THANK
YOU!